

Test 3: Convergence rate (power-law fit)

Verdict: PASS | Generated: 2026-09-02 13:48:50 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

log-log regression of $b(N)$: the slope is a formula-derived quantity with R^2 as the goodness yardstick.

METHOD: log-log regression of $b(N)$: the slope is a formula-derived quantity with R^2 as the goodness yardstick.

Result

Test 3 [HARDCORE pass 2]: 4 sub-checks, 0 failed, $b(50000)=1.2126$? PASS

``plots/plot_01` (.png 600dpi / .svg / .pdf)`

Plots

- ``plots/animation.gif`` ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v19.jl --test 3 --no-two-pass`

Independent verification guide

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(no `log_comp` entries)

FULL COMPUTATION LOG

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CONSOLE CAPTURE

TEST 3 (secondary, HARDCORE): Objection 1: $b(N)$ Convergence ? deep rate-law validation

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R1 dense power law: $b(N)$? $7.3564 \cdot N^{(-0.1737)}$, $R^2 = 0.988236$ (limit 0.95)

R2 alt $1/\log N$: $b(N)$? $-0.2557 + 16.2185/\log(N)$, $R^2 = 0.997500$

best of two laws: $R^2 = 0.9975$ (limit 0.95)

R3 walk-forward : median err 0.05%, max err 0.09% (limits 2%/5%)

R4 ? stability : $?_{\text{lower}}=0.2116$ vs $?_{\text{upper}}=0.1393$, $|??|=0.0723$ (limit 0.2)

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R1 power-law R^2 PASS ? $R^2 = 0.9882$

R2 best-law R^2 PASS ? $R^2 = 0.9975$

R3 walk-forward PASS ? median 0.05%, max 0.09%

R4 ? stability PASS ? $|??| = 0.0723$

Test 3 [HARDCORE pass 2]: 4 sub-checks, 0 failed, $b(50000)=1.2126$? PASS